System Architecture & Design

Technical Reference Specification

> *Sample publication demonstrating core markpublish features: Typst typesetting, mathematical formulas, syntax highlighting, and callout boxes.*

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1 Introduction & System Architecture

markpublish transforms structured Markdown files into publication-grade documents. Structure, metadata, and styling are governed centrally in `markpublish.yaml` — everything else remains clean prose.

> **Note**
> The native Typst typesetting engine delivers book-grade printing, automatic hyphenation, and consistent typography without external LaTeX or Pandoc chains.

1.1 Modular Structure

Documents can be organized into parts, chapters, and divider pages. Each file remains pure Markdown, while the central manifest handles chapter ordering and automatic numbering.

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2 Figures & References

A graphic with a caption becomes a numbered figure. markpublish sets the word and number itself, and every figure can be referenced in the text by its ID.

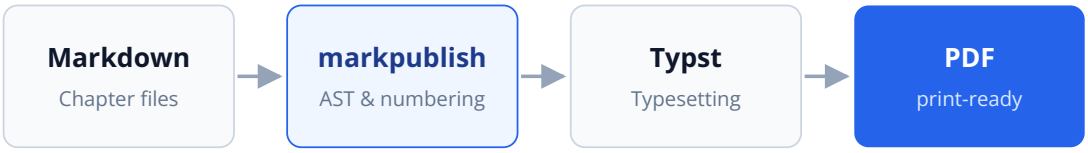


Figure 1: From Markdown chapter to print-ready PDF

As Figure 1 shows, markpublish reads the chapter files, numbers headings, figures and tables, and hands the result to Typst. References like this one always stay correctly numbered and are clickable in the PDF.

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3 Tables & Captions

Tables carry their caption on top, as is customary in book typesetting. Numbering and references work just like they do for figures.

Table 1: Captioned elements and how to write them

Element	Markdown	Result in the PDF
Figure	`/// figure-caption`	numbered, in the list of figures
Table	`/// table-caption`	numbered, caption on top
Reference	`[](#id)`	“Figure 1”, clickable
List	`list_of: figures`	a page of its own in the document

Table 1 sums up the notation. The words “Figure” and “Table” come from the i18n cascade and thus follow the document language.

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4 Scientific Computations

Mathematical formulas are typeset with high typographic precision using the native Typst math engine:

$$E = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

4.1 Pipeline Implementation

Source code blocks benefit from syntax highlighting with bundled fonts:

```
# pipeline.py
from pathlib import Path
import typst

def render_pdf(manifest_file: Path) -> bytes:
    """Compiles the document via Typst AST."""
    doc = load_manifest(manifest_file)
    return typst.compile(doc.to_typst())
```

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5 Notes & Checklist

Critical information stands out with GitHub-style callouts:

> **Tip**
> A unified numbering pattern like `1|.1|+` synchronizes headings and tables of contents automatically.

> **Warning**
> Custom themes should always be validated locally using `markpublish build` prior to release.

5.1 Publication Checklist

- [x] Validate project manifest `markpublish.yaml`
- [x] Compile via Typst engine
- [x] Verify accessible PDF standards
- [ ] Sign off publication

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